ADICNESS OF THE SECOND OF TWO COMPOSABLE RING MAPS IS PERMANENT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Ideal of definition, pre-adic topological ring, and adic ring homomorphism. Same chapter, Definition definition-adic-homomorphism (source lines 1491-1495) together with the surrounding discussion of pre-adic rings. Fixes the notion that has to be produced for B -> C and the vocabulary of ideals of definition used throughout.
- An ideal of definition is determined by a comparison with any other ideal of definition. Same chapter, the discussion following Definition definition-weakly-adic, where ideals of definition are seen to form a fundamental system of open ideals. Justifies the final step: an ideal sandwiched between two ideals of definition is itself an ideal of definition.

Lemma 0.1. *Let $A \to B$ and $B \to C$ be continuous homomorphisms of pre-adic rings. If $A \to C$ is adic, then $B \to C$ is adic.*

Proof. Choose an ideal of definition I of A. As $A \to C$ is adic, we see that IC is an ideal of definition of C. As $B \to C$ is continuous, we can find an ideal of definition $J \subset B$ mapping into IC. As $A \to B$ is continuous the inverse image $I' \subset I$ of J in I is an ideal of definition of A too. Hence $I'C \subset JC \subset IC$ is sandwiched between two ideals of definition, hence is an ideal of definition itself. □

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